The Hidden Toll: Public Health Burden of Illicit Opioid Market Shifts

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This paper estimates the causal impact of a major supply-side shock in the U.S. illicit opioid market. Beginning in 2012, Mexican drug trafficking organizations displaced Colombian suppliers and took control of white-powder heroin distribution in the Eastern U.S. This transition raised the mean and variance of heroin purity and facilitated widespread fentanyl adulteration. Using 2004–2019 discharge-level data from the National Inpatient Sample and an event-study design, we show that this shift led to a 73 percent (0.11 percentage point) increase in heroin overdose inpatient admissions in affected regions. Placebo tests using alcohol poisoning and cocaine show no comparable effects, while spillovers to fentanyl- and methadone-related admissions are positive but imprecise. The effects are concentrated among Medicaid-covered and self-pay patients, indicating substantial public sector exposure. We estimate direct medical costs of approximately $4 billion, but the total welfare loss is far larger once accounting for excess mortality, reduced labor force participation, and intergenerational deficits in human capital.

The opioid[1] epidemic represents one of the most devastating public health crises in modern U.S. history (Cutler and Glaeser, 2021). It has unfolded in successive waves shaped by shifts in medical practice (Dowell et al., 2016; Arteaga and Barone, 2025), regulatory reforms (Evans et al., 2019), and illicit supply chains (Ciccarone, 2019). Earlier phases were fueled by prescription opioids, while the current and most lethal phase is driven by heroin and synthetic opioids, particularly fentanyl, which is especially dangerous because it often enters the illicit market through undetectable adulteration (Ciccarone et al., 2017; CDC, 2024; Donahoe and Soliman, 2025). Yet beyond national aggregates, market-level variation remains understudied, even though geographic segmentation could create sharp differences in overdose risk and welfare losses.

We exploit one such episode: the 2012 displacement of Colombian suppliers by Mexican drug trafficking organizations (DTOs) in the Eastern U.S. This transition marked a structural break in the heroin supply chain, and the new white powder heroin supplied by

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[1] Opioids act on the central nervous system to relieve pain and induce euphoria. They include prescription medications and illicit substances with high addictive potential and overdose risk. Opioids are commonly categorized into three groups: natural opioids (opiates) derived from the opium poppy (e.g., morphine, codeine); semi-synthetic opioids, chemically modified from natural opiates (e.g., heroin, oxycodone); and fully synthetic opioids, manufactured from chemical precursors without natural sources (e.g., fentanyl, tramadol) (NIDA, 2024).

Mexican DTOs differed in two key ways. First, both the mean purity and the variance of purity increased, raising the probability of extremely high-potency exposures.[2] Second, unlike black tar heroin, which long dominated Western markets, Mexican white powder heroin was chemically and physically similar to fentanyl, enabling seamless blending and widespread adulteration (DEA, 2015a, 2019). Users in eastern markets, therefore, faced increased overdose risk from more variable and fentanyl-contaminated heroin, often unknowingly. These quality shifts coincided with sharp increases in overdose deaths (Donahoe and Soliman, 2025).

Most overdoses, however, are nonfatal: between 2010 and 2020, roughly 85 percent of opioid-involved overdoses did not result in death (Casillas et al., 2024). In this study, we combine hospital discharge data from the 2004–2019 National Inpatient Sample with an event study framework to estimate the effect of this heroin supply shock on inpatient overdose admissions. Exploiting the longstanding geographic segmentation of the heroin market along the Mississippi River, we define treatment as residing in previously Colombian-supplied regions but exposed to the Mexican takeover beginning in 2012. We estimate that heroin-related inpatient admissions increased by 0.11 percentage points in affected regions, corresponding to a 73 percent rise relative to the pre-shock mean. The effect is concentrated among Medicaid-covered and self-pay patients, implying substantial fiscal exposure for state and federal programs. Placebo tests using alcohol- and cocaine-related admissions, together with robustness checks, support the validity of our identification strategy.

We then quantify the welfare implications. We estimate direct costs of approximately $4 billion, including excess inpatient and emergency care, permanent disability, and medication-assisted treatment. Beyond these direct costs, the shock likely imposed much larger indirect burdens that are difficult to monetize: excess mortality (Florence et al., 2016; Lewer et al., 2021), reduced labor force participation and increased reliance on disability programs (Krueger, 2017; Harris et al., 2020), and intergenerational human capital deficits through foster care placements and reduced educational attainment (Quast et al., 2019; Meinhofer et al., 2024). Taken together, these findings demonstrate fiscal and institutional consequences of illicit supply disruptions and underline the underprovision of surveillance and harm-reduction infrastructure as a classic public goods failure.

Although our setting is the U.S. opioid crisis, the lessons are general. This episode shows how illicit supply shocks propagate through health systems, labor markets, and communities; offers a cautionary tale for emerging synthetic drug markets worldwide (e.g., methamphetamine in Asia, tramadol in Africa, fentanyl analogs globally); and underscores the economic importance of surveillance in markets plagued by asymmetric information and externalities. More broadly, our findings contribute to a growing literature on how shifts in product composition and supply chain control in illicit markets shape economic outcomes.

[2]Drug Enforcement Administration (DEA) laboratory tests showed that the new Mexican heroin has fatter upper tails in the distribution of purity (DEA, 2015b, 2016a, 2017).

I. Background

Donahoe and Soliman (2025) provide an overview of the evolution of the U.S. illicit opioid market, particularly emphasizing the shift in heroin supply from Colombian to Mexican DTOs around 2012. We extend their analysis by documenting additional institutional details on historical market segmentation, the persistence and underlying causes of user preferences, and a more precise timeline of the supply-side transition. These extensions sharpen the quasi-random geographic variation in exposure to opioid quality shocks and thereby strengthen the validity of our empirical identification strategy.

A. *User Preferences and Market Persistence (Demand Side)*

Heroin and fentanyl[3] are particularly consequential in illicit opioid markets.[4] Heroin, a semi-synthetic opioid, has historically circulated in two distinct forms: black tar heroin (BTH) and white powder heroin (WPH). Although chemically similar, these forms differ substantially in production methods, purity, and consumer practices.

The initial divergence stemmed from supply-side production characteristics. Black tar heroin requires minimal purification and fewer specialized chemical precursors, aligning with the local poppy strains cultivated in Mexico and the capabilities of small, dispersed laboratories (United Nations Office on Drugs and Crime, 2010). Mexican DTOs naturally favored BTH production due to its lower manufacturing costs, simplified chemical processing methods, and suitability for decentralized operations. Conversely, Southeast Asian and later Colombian DTOs developed more advanced processing techniques and laboratory infrastructures to refine white powder heroin. WPH required specific precursors and rigorous purification steps, but its high solubility and chemical purity made it particularly attractive in markets favoring snorting—practices especially prevalent along the East Coast of the U.S. (Ciccarone, 2009; Mars et al., 2016; National Institute on Drug Abuse, 2021).

These production-based differences also shaped distinct cultural and social perceptions across regions. Black tar heroin, known for its sticky, resinous texture and low solubility, is typically administered intravenously, a method that increases the risk of vein damage and transmission of infectious diseases such as HIV and Hepatitis C (Ciccarone and Bourgois, 2003). Predominantly available in the Western U.S., BTH has long been stigmatized as a lower-quality and harsher product, associated with marginalized and lower-income populations (Mars et al., 2016). In contrast, white powder heroin, characterized by its fine granularity and high solubility, could be easily snorted.[5] This, in turn, gives WPH a more refined reputation and was regarded as more socially acceptable (Frank, 2000;

[3]Fentanyl, a fully synthetic opioid initially developed for medical pain relief, is approximately 20–50 times more potent than heroin. Despite becoming a primary driver of the current opioid crisis, fentanyl itself is generally unpopular among traditional heroin users due to its unpredictability and high overdose risk. However, its low production cost has led to widespread use as an adulterant in heroin and counterfeit pills.

[4]By 2023, fentanyl had surpassed heroin as the leading opioid in U.S. illicit markets, implicated in nearly 70% of opioid-related overdose deaths. At the same time, the share of heroin in the illicit market steadily declined due to fentanyl substitution and adulteration (CDC, 2023; DEA, 2024).

[5]WPH can also be injected, but unlike BTH, heating is unnecessary.

Mars and Ciccarone, 2015). These cultural distinctions reinforced regional segmentation and contributed to the stickiness of user preferences.

Over time, this interaction between supply conditions and consumer practices created path-dependent market structures. Mexican DTOs repeatedly attempted to introduce black tar heroin into Eastern markets during the early 2000s but failed due to entrenched consumer preferences, practical barriers to use, and cultural stigma (DEA, 2015a; Ciccarone, 2009; Mars et al., 2016).[6] As a result, regional consumption habits stabilized, locking in distribution networks optimized for each heroin form. This persistence provides a clear framework for understanding why later exogenous supply shocks had such asymmetric impacts across U.S. regions.

B. Historical Evolution of the U.S. Heroin Market (Supply Side)

Pre-2012 Era

Despite stable demand-side preferences, the U.S. heroin market underwent major transformations on the supply side, transitioning from a fragmented, regionally segmented structure into an increasingly integrated and dangerous national market in the early 21st century.

Before the 1990s, U.S. heroin markets were characterized by relatively low purity and high cost, which limited consumption primarily to intravenous injection. At that time, Asian heroin, primarily from the "Golden Triangle", dominated East Coast markets. These products are typically entered through West Coast ports before being distributed across the country. Concurrently, Mexican-sourced black tar heroin was predominant in the western U.S., benefiting from geographic proximity and established cross-border trafficking infrastructure (NDIC, 2006).

In the early 1990s, the entrance of Colombian DTOs marked a fundamental shift. Leveraging their extensive cocaine trafficking networks, Colombian DTOs introduced high-purity, competitively priced white powder heroin to Eastern U.S. markets. This competitive advantage quickly displaced Asian suppliers, establishing Colombia as the primary supplier for the eastern U.S. heroin market by the late 1990s (Rosenblum et al., 2014).

By the early 2000s, regional segmentation had solidified, with Colombian DTOs controlling white powder heroin in the East and Mexican DTOs remained dominant in black tar heroin in the West. A transitional zone emerged in the Midwest, with the Mississippi River serving as an approximate boundary. The top panel of Figure 1 presents this pre-2012 segmentation: white powder heroin accounted for a majority share of exhibits (DEA undercover purchases of street-level samples) in Eastern cities but remained almost absent in the West, where black tar heroin dominated.[7]

This equilibrium was gradually destabilized by two major events. First, the Colombian government's increased extradition of narcotics traffickers to the United States in the late 1990s severely weakened the operational capacity of Colombian DTOs (Pace Intl. Law

[6] Ciccarone and Bourgois (2003) describes this as a specialized form of drug "connoisseurship."
[7] See also Ciccarone and Bourgois (2003); DEA (2015a); Donahoe and Soliman (2025).

Review, 2012). Second, the post-9/11 escalation in U.S. airport security substantially raised the cost and risk of air trafficking, disrupting Colombian heroin export routes (Donahoe and Soliman, 2025).

The 2012 Takeover and Opioid Quality Shock

As Colombian DTOs lost distribution capacity, they increasingly shifted toward wholesaling heroin to Mexican DTOs, thereby reducing direct conflict with the DEA and lowering enforcement risks. Mexican DTOs capitalized on this disruption by leveraging established cross-border smuggling networks to penetrate Eastern heroin markets and by adopting advanced processing techniques to produce refined white powder heroin tailored to regional preferences. By 2013, DEA chemical analyses first detected white powder heroin in Eastern markets with signatures consistent with Mexican production, and the share of suspected Mexican-origin product increased rapidly in the following years. By 2015, the DEA's Heroin Signature Program (HSP) identified that 93 percent of wholesale-level heroin samples originated from Mexico, with the remaining 7 percent sourced from South America, Southwest Asia, or of unidentified origin.

The bottom panel of Figure 1 illustrates this post-2012 transition, as an increasing share of Eastern white powder heroin exhibits were either suspected or verified to originate from Mexican DTOs, coinciding with the documented erosion of Colombian dominance.[8] This shift effectively dissolved the long-standing regional segmentation of the U.S. heroin market and consolidated control of both major heroin forms under Mexican DTOs (DEA, 2016b; Donahoe and Soliman, 2025).[9]

This takeover also fundamentally altered the risk profile of the Eastern U.S. heroin market by increasing average purity and introducing substantial variability in potency. In 2015, the DEA formally introduced a new classification (MEX-SA) to designate heroin of Mexican origin processed using South American (primarily Colombian) methods. Samples in this category exhibited the highest average purity at 70%, followed by South American heroin at 63% and Mexican black tar heroin at approximately 40% (DEA, 2015c, 2018a).[10]

In addition to high purity, Mexican white powder heroin displayed significantly greater variability in composition compared to Colombian heroin. Colombian heroin was typically adulterated near its production source, yielding relatively consistent potency, whereas Mexican white powder heroin was often cut further downstream at regional hubs or local distribution points (DEA, 2018b; Montero et al., 2022).[11] This fragmentation of the supply chain introduced substantial inconsistency in product strength and

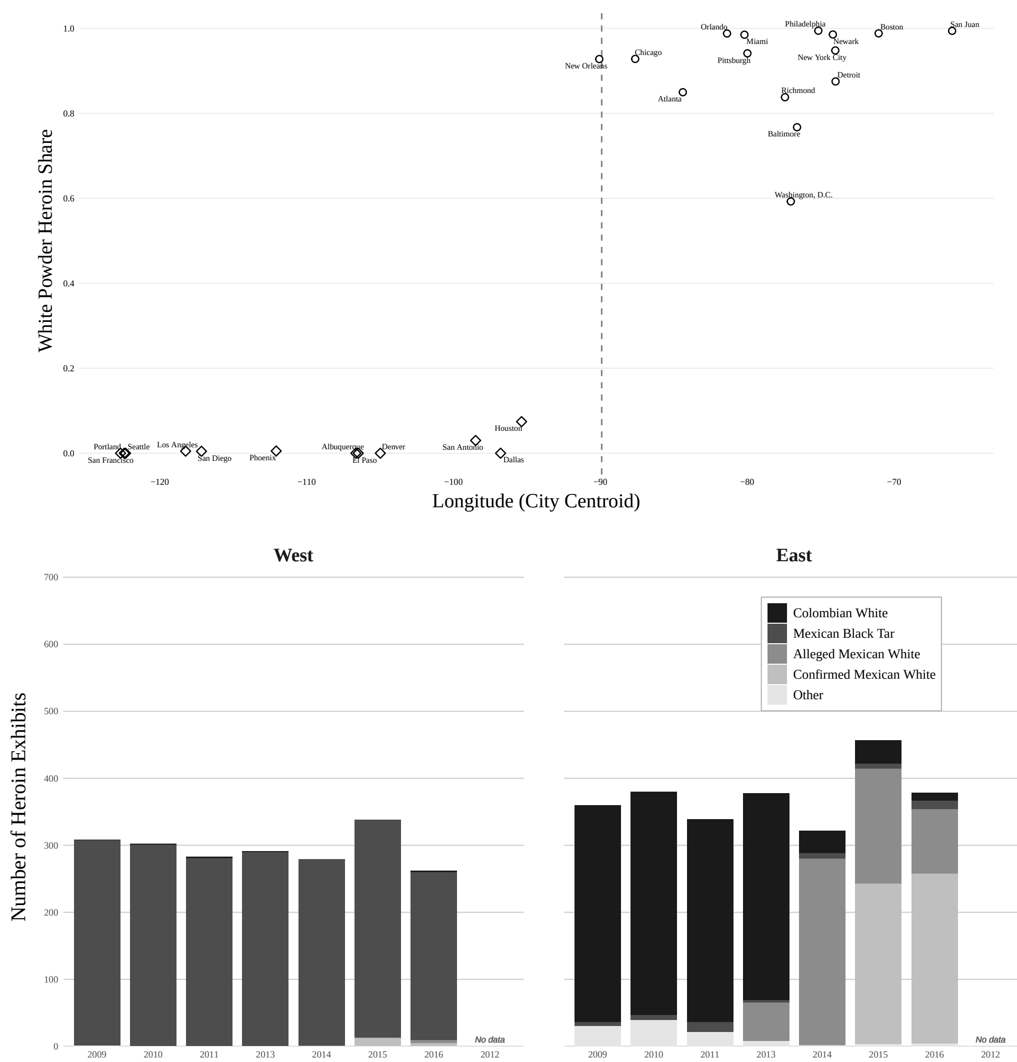
[8] This figure aligns closely with Donahoe and Soliman (2025), as both are based on the HDMP. However, ours excludes St. Louis and Minneapolis, which lie along the Mississippi River and have historically been supplied by both Mexican and Colombian DTOs, potentially generating ambiguous classification with respect to the supply shock.

[9] See long-run heroin market share dynamics in Figure A5.

[10] Average purity estimates vary across DEA reports depending on whether the analysis is conducted at the wholesale or retail level, due to post-production adulteration and dilution. Adulterants refer to pharmacologically active substances added to mimic or enhance heroin's effects (e.g., fentanyl), while diluents are inactive substances, such as lactose, mannitol, starch, or sucrose, used to increase volume.

[11] Mexican DTOs mainly serve as wholesale suppliers and depend on U.S.-based street-level gangs for drug distribution to distance risk (DEA, 2015a).

Figure 1. Geographic Distribution and Temporal Trends in Heroin Origin



Notes: The **top panel** plots the share of heroin exhibits identified as white powder heroin (South American origin) against the longitude of each city's centroid, based on data from the DEA's Heroin Domestic Monitor Program. The sample includes major U.S. cities reporting to the HDMP between 2004 and 2011, excluding Minneapolis and St. Louis because their Census division was excluded from the analysis sample. Western cities are indicated by open diamonds and Eastern cities by open circles. The dashed vertical line at longitude –90 approximates the Mississippi River, which is used as the geographic demarcation between the East and West heroin markets. City labels are positioned to avoid overlap. The **bottom panel** shows the number of heroin exhibits by origin type for East and West regions from 2009 to 2016. As part of the HDMP, most participating cities conduct ten street-level heroin purchases per quarter (40 annually). In New York City, a major heroin distribution hub, this target is 15 purchases per quarter (60 annually). A smaller subset of cities, including Albuquerque (NM), Houston (TX), Orlando (FL), Pittsburgh (PA), Portland (OR), Richmond (VA), and San Antonio (TX), is assigned only five purchases per quarter (20 annually). In total, the HDMP scheduled 960 heroin exhibits for purchase nationwide in 2016 (DEA, 2018a).

heightened overdose risks.

Another driver of the increased volatility was the growing use of fentanyl as an adulterant (Ciccarone, 2009; Mars et al., 2016). Despite its potency, fentanyl is generally unappealing to many opioid users due to its short-lasting high and significantly higher mortality risk. Most heroin users prefer the longer-lasting effects of heroin and seek to avoid the dangers associated with fentanyl. However, fentanyl is most often consumed unknowingly, mixed into heroin without the user's awareness (DEA, 2015a). Between 2015 and 2016, the number of heroin exhibits analyzed by the DEA's Heroin Domestic Monitor Program (HDMP) that tested positive for fentanyl or its analogs nearly tripled: from 53 to 158 (DEA, 2018a).

Fentanyl's chemical and physical similarity to powdered heroin made it an ideal, though exceptionally hazardous, cutting agent.[12] Its substantially lower cost relative to traditional opioids further incentivized its widespread use across fragmented and poorly regulated downstream distribution networks. As a result, fentanyl adulteration became particularly prevalent in Eastern markets historically dominated by WPH,[13] where its presence often went undetected by users and even street-level dealers (DEA, 2016b; Donahoe and Soliman, 2025).

Conversely, black tar heroin markets in the Western U.S., which remained under Mexican DTOs' control throughout, experienced minimal fentanyl adulteration. The sticky and resinous texture of black tar heroin made it difficult to mix with powdered fentanyl. Moreover, black tar heroin typically requires heating prior to injection, which likely causes fentanyl to vaporize before use. These physical and pharmacological barriers significantly limited the incorporation of fentanyl into Western markets (DEA, 2018a).

Taken together, these developments reveal a stark asymmetry in the evolution of heroin markets across U.S. regions since 2012. Eastern markets experienced a profound supply-side shock marked by new suppliers, altered drug formulations, and a sharp rise in overdose risk. Meanwhile, Western markets remained comparatively stable. The decline of Colombian DTOs not only marked a transition in supply chain dominance but also ushered in a more dangerous era of decentralized fentanyl adulteration—one that continues to define the current phase of the U.S. opioid crisis.

II. Data

Our primary data source is the *National Inpatient Sample (NIS)*, the largest publicly available, all-payer inpatient database in the United States. We use annual discharge-level records from 2004–2019, covering a stratified sample of approximately 20 percent of U.S. community hospital stays. The NIS, maintained by the Healthcare Cost and Utilization Project (HCUP) of the Agency for Healthcare Research and Quality, provides detailed information on patient demographics, diagnoses, procedures, payer type, and hospital characteristics, along with discharge weights that permit national inference (Agency for Healthcare Research and Quality, Accessed 2025).

[12]Even microgram-level errors in fentanyl blending can result in fatal potency variations (Ciccarone, 2017).

[13]Fentanyl almost exclusively adulterated into WPH instead of other white power drugs (DEA, 2019).

We identify drug-related inpatient admissions using International Classification of Diseases (ICD) diagnosis codes. Substance categories include heroin, fentanyl (synthetic opioids other than methadone), methadone, cocaine, and alcohol poisoning. For each substance, we construct yearly, patient-level indicators for having any related diagnosis. We link each patient's hospital state to its Census division[14] and classify states as east or west of the Mississippi River to define treatment and control regions. Individual-level covariates include age, sex, race/ethnicity, urban–rural residence, and quartile of income. Table A1 presents summary statistics of demographics for treatment and control groups.

A challenge for longitudinal consistency is the 2015 transition from ICD-9 to ICD-10 codes. To mitigate concerns, we draw on the Centers for Medicare & Medicaid Services General Equivalence Mappings, distributed via the National Bureau of Economic Research crosswalk files (NBER, Accessed 2025), which provide concordances between ICD-9-CM and ICD-10-CM/PCS codes. We exclude 2015 from the baseline analysis, since NIS itself reports incomplete coverage during the transition year, but verify that outcome trends remain stable in surrounding years.

III. Empirical Strategy

Our empirical strategy exploits the abrupt and geographically concentrated supply-side shock to the U.S. heroin market in 2012, when Mexican DTOs displaced Colombian suppliers in Eastern states and introduced a new form of white powder heroin with higher and more variable potency. As detailed in Section I, this transition increased overdose risk in markets historically dependent on Colombian heroin, while Western markets remained largely unaffected. The persistence of pre-existing market segmentation, shaped by entrenched user preferences and distribution networks, creates a natural experiment for estimating the causal effects of this shock on overdose-related hospitalizations.

We implement two complementary specifications. Our primary design is an event-study model estimated at the discharge level:[15]

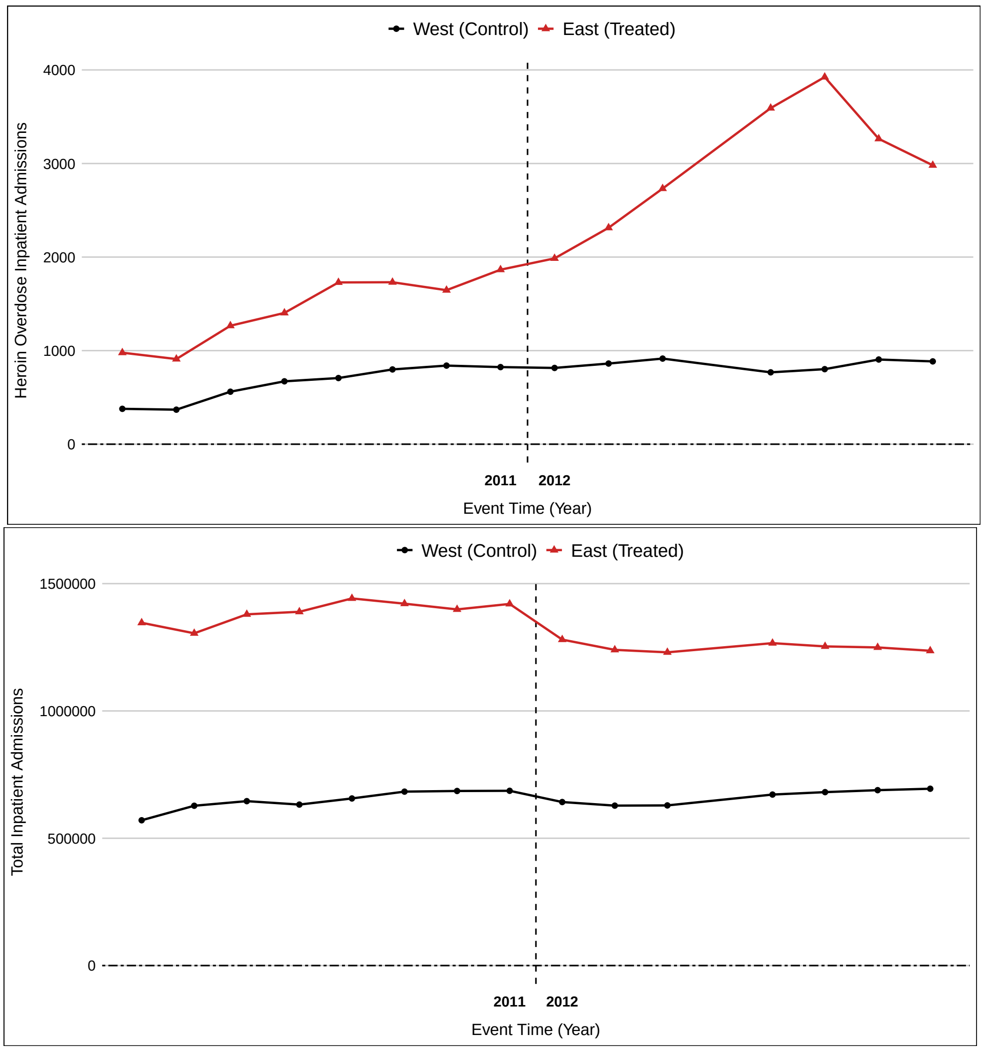
$$Y_{idt} = \sum_{\tau \neq 2011} \beta_\tau \cdot \mathbb{1}(t = \tau) \times \text{East}_d + \alpha_d + \lambda_t + \gamma' X_{idt} + \varepsilon_{idt} \tag{1}$$

where Y_{idt} denotes a binary indicator for whether discharge i in division d and year t is an inpatient admission with a substance-specific overdose/poisoning (e.g., heroin, fentanyl, cocaine, methadone, or alcohol). East_d is an indicator equal to one for treatment divisions east of the Mississippi River, while $\mathbb{1}(t = \tau)$ is a set of event-time dummies, where 2012 is the event year and 2011 is omitted as the reference year. The vector X_{idt} includes individual-level characteristics (age, sex, race/ethnicity, urban-rural residence, insurance payer type, and ZIP-code income quartile). Division fixed effects (α_d) absorb time-invariant regional differences, while year fixed effects (λ_t) capture common shocks.

[14]The NIS identifies hospitals only at the level of the nine Census divisions. We exclude the West North Central and East South Central divisions because several major cities in these regions have historically been supplied by both Mexican and Colombian DTOs, given their proximity to the Mississippi River.

[15]Following (Donahoe and Soliman, 2025).

Figure 2. Inpatient Admissions by Region: Heroin-Related and Total



Notes: **Top panel** plots heroin-related inpatient admissions for treatment (East) and control (West) regions by calendar year. **Bottom panel** plots total inpatient admissions for the same regions. In both panels, points and lines denote annual series by region, and the dashed vertical line between 2011 and 2012 marks the pre/post boundary relative to the heroin supply shock. Data are from the NIS, covering 2004–2019 (excluding 2015).

We use NIS discharge weights to ensure national representativeness. Standard errors are clustered at the division level. Given the limited number of clusters, we also report wild-cluster bootstrap standard errors. The coefficients β_τ measure dynamic treatment effects relative to 2011: pre-shock estimates test for parallel trend, while post-shock estimates trace the magnitude and persistence of the supply-shock effects.

Figure 2 plots raw trends in inpatient admissions. The bottom panel shows that total inpatient admissions moved in parallel in the East and the West, indicating no differential shifts in overall admission levels. In contrast, the top panel shows that heroin overdose–related admissions diverged after 2011: admissions increased sharply in the East while remaining stable in the West. This divergence aligns with the timing and geographic scope of the heroin supply shock.[16]

As a complementary specification, we estimate a two-way fixed effects Difference-in-Differences (DiD) model that captures the average post-shock treatment effect:

$$Y_{idt} = \beta \cdot \text{Post}_t \times \text{East}_d + \alpha_d + \lambda_t + \gamma' X_{idt} + \varepsilon_{idt} \tag{2}$$

where Post_t equals one in years after 2011. The coefficient β captures the average difference in outcomes between the East and the West in the post-shock period relative to the pre-shock baseline. The specification includes the same covariates and fixed effects as in the event-study model.

The two specifications are complementary. The event-study shows the time path of treatment effects and provides a direct test of pre-trends, while the two-way fixed-effects model recovers an interpretable average effect that can be linked to welfare analysis. Causal interpretation of these estimates requires that, absent the 2012 heroin supply shock, heroin-related inpatient admissions in the East and the West would have followed similar trends. This assumption is supported by the event-study, which shows no differential pre-trends, and by placebo outcomes in alcohol and cocaine, which show no treatment effects. The design further requires that no other contemporaneous shocks differentially affected Eastern markets and that spillovers into Western markets were limited. Under these conditions, the DiD specifications identify the causal effect of the 2012 heroin supply shock on heroin overdose inpatient admissions.

IV. Results

A. Difference-in-Differences Estimates

Table 1 reports the baseline DiD estimates of the 2012 heroin market shock on the probability of heroin-related inpatient admissions. Across specifications, the estimated effect is positive, stable, and statistically significant at the 1 percent level. In our preferred specification (column 4), which includes demographic controls and discharge weights, the takeover increased the probability of heroin overdose inpatient admission by 0.11 percentage points (robust SE = 0.0002, bootstrap SE = 0.0002). Relative to the pre-shock

[16]Figure A3 plots the share of heroin overdose admissions for another reference of the following event study.

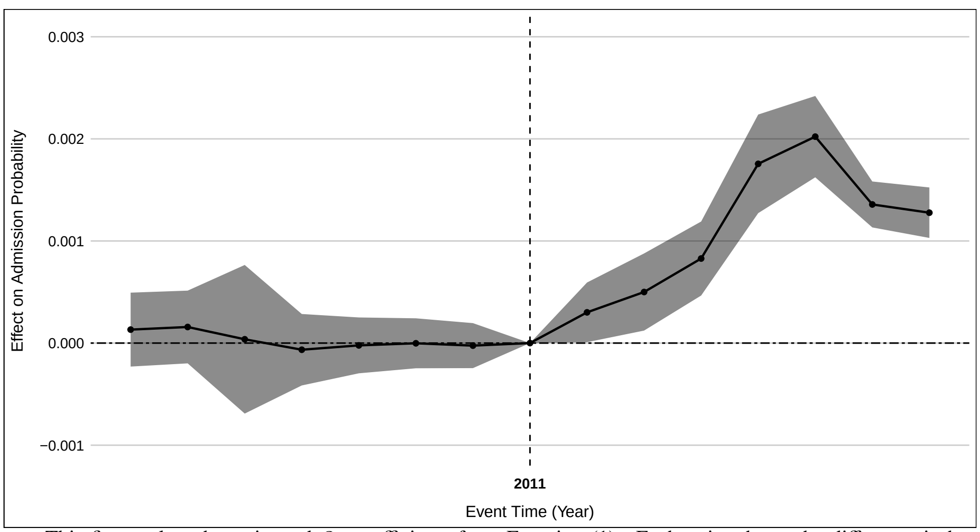
Table 1— DiD Estimates: Heroin Overdose Inpatient Admissions

	(1)	(2)	(3)	(4)	(5)
Post × Treated	0.0010***	0.0010***	0.0011***	0.0011***	0.0013***
	(0.0002)	(0.0002)	(0.0002)	(0.0002)	(0.0002)
	[0.0002]	[0.0002]	[0.0002]	[0.0002]	[0.0002]
Baseline	0.0015	0.0015	0.0015	0.0015	0.0015
Observations	29,680,496	29,680,496	25,146,035	25,146,035	23,344,341
Controls	✗	✗	✓	✓	✓
Weights	✗	✓	✗	✓	✓

Notes: This table reports DiD estimates of the effect of the 2012 heroin supply shock on the probability of heroin overdose inpatient admissions. The dependent variable is a binary indicator for heroin overdose inpatient admission. The treatment group includes regions east of the Mississippi River; the control group includes regions west. "Post" denotes the post-2011 period. All specifications include Census division and calendar year fixed effects. Where indicated, models also include individual-level demographic controls and discharge weights. Column (5) excludes 2012, when markets were in transition. Standard errors clustered at the census division level are reported in parentheses. Bootstrap standard errors are shown in square brackets.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 3. Event Study: Heroin Overdose Admissions



Notes: This figure plots the estimated β_τ coefficients from Equation (1). Each point shows the difference in heroin overdose inpatient admission rates between treatment and control regions relative to the reference period (2011), which immediately precedes the supply shock. The event time spans from 2004 to 2019, excluding 2015 due to data unavailability. All specifications include Census division and calendar year fixed effects, as well as individual-level demographic controls and discharge weights.

mean of 0.15 percent, this represents a 73 percent increase attributable to the supply shock. Point estimates across alternative specifications remain in the 0.10–0.13 percentage point range, which indicates that neither compositional changes in patient characteristics nor weighting choices drive the results.

The corresponding event-study estimates in Figure 3 show no detectable pre-treatment differences between East and West, supporting the parallel trends assumption. Beginning in 2012, the effect rises sharply, peaking around 0.20 percentage points above baseline by 2016–2017, and remains elevated through 2019. This persistence suggests that the supply shock produced a maintained increase in overdose risk rather than a short-term adjustment. Using the NIS sampling weights, our preferred estimate implies roughly 27,000 additional heroin overdose inpatient admissions annually in the East between 2012 and 2019, about 215,000 excess admissions over the eight-year post shock, which provides the foundation for our welfare cost estimates in Section VI.

B. Heterogeneity by Subgroup

Race/Ethnicity

Table 2— DiD Estimates: Heroin Overdose Inpatient Admissions by Race

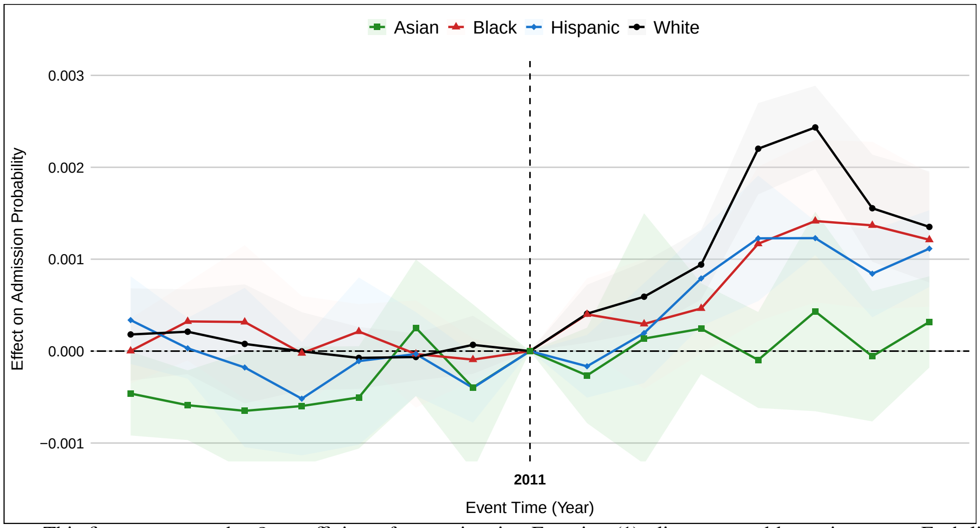
	(1) White	(2) Black	(3) Hispanic	(4) Asian
Post × Treated	0.0013***	0.0008**	0.0009**	0.0004**
	(0.0002)	(0.0002)	(0.0003)	(0.0002)
	[0.0002]	[0.0002]	[0.0003]	[0.0002]
Baseline	0.0017	0.0011	0.0011	0.0006
Observations	15,100,797	5,197,312	3,311,752	537,690
Controls	✓	✓	✓	✓
Weights	✓	✓	✓	✓

Notes: This table reports DiD estimates of the effect of the 2012 heroin supply shock on inpatient admissions for heroin-related overdoses, disaggregated by race/ethnicity. Each column presents a separate regression. The treatment group includes regions east of the Mississippi River; the control group includes regions west. "Post" denotes the post-2011 period. All regressions include Census division and calendar year fixed effects, as well as individual-level demographic controls and discharge weights. Standard errors clustered at the census division level are reported in parentheses. Bootstrap standard errors are shown in square brackets.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 2 and Figure 4 present DiD estimates disaggregated by patient race/ethnicity. The 2012 heroin supply shock increased heroin overdose admissions for all racial groups, but the magnitudes vary substantially. The largest absolute and relative increase occurred among White patients: 0.13 percentage points (SE = 0.0002), or 76 percent above their baseline mean of 0.17 percent. Black and Hispanic patients experienced increases of 0.08 and 0.09 percentage points, 73 and 82 percent above their baselines. Asian patients saw

Figure 4. Event Study by Race: Heroin Overdose Admissions



Notes: This figure presents the β_τ coefficients from estimating Equation (1), disaggregated by patient race. Each line shows the differential change in heroin-related inpatient admissions between treatment and control regions relative to the reference period (2011), which immediately precedes the supply shock. Event time spans from 2004 to 2019, excluding 2015 due to data unavailability. All specifications include Census division and calendar-year fixed effects, as well as individual-level demographic controls and discharge weights.

a smaller increase of 0.04 percentage points (67 percent above baseline). All effects are statistically significant at the 5 percent level.

The event study estimates in Figure 4 show parallel pre-trends across racial groups and a sharp, persistent divergence post shock, with the steepest rise among White patients. This pattern is consistent with prior evidence that White populations were more heavily exposed to the prescription opioid epidemic and transitioned into heroin use following tighter prescribing regulations (Cicero et al., 2014; Unick et al., 2013). In contrast, Black Americans were less exposed to prescription opioids due to systemic under-prescribing and discrimination in pain management (Pletcher et al., 2008; Hoffman et al., 2016); Hispanic communities historically displayed higher cocaine use relative to heroin (Kilmer et al., 2010); and Asian Americans often prefer methamphetamine (Substance Abuse and Mental Health Services Administration, 2021).

This heterogeneity highlights how structural racism in health care created unequal exposure across groups. Discriminatory under-prescribing insulated many Black patients from the initial prescription wave, while over-prescribing to White patients created a large pool of individuals vulnerable to transitioning into heroin markets. The 2012 supply shock therefore, disproportionately magnified risks for White patients, bridging pre-existing disparities in prescription opioid exposure into the illicit market. This redistribution of risk underscores a broader policy lesson: safeguards applied unevenly across populations may reduce overall access but inadvertently reallocate harms across racial groups and drug types.

Insurance Coverage

Table 3— DiD Estimates: Heroin Overdose Inpatient Admissions by Payer Type

	(1) Medicare	(2) Medicaid	(3) Private	(4) Self-Pay
Post × Treated	0.0006**	0.0020***	0.0007***	0.0018***
	(0.0002)	(0.0005)	(0.0001)	(0.0002)
	[0.0002]	[0.0005]	[0.0001]	[0.0002]
Baseline	0.0010	0.0024	0.0009	0.0026
Observations	5,406,831	5,974,155	9,518,510	2,699,185
Controls	✓	✓	✓	✓
Weights	✓	✓	✓	✓

Notes: This table reports DiD estimates of the effect of the 2012 heroin supply shock on inpatient admissions for heroin-related overdoses, disaggregated by primary payer type. Each column presents a separate regression for Medicare, Medicaid, Private Insurance, and Self-Pay, respectively. All models include Census division and calendar year fixed effects, as well as individual-level demographic controls and discharge weights. Standard errors clustered at the census division level are reported in parentheses. Bootstrap standard errors are shown in square brackets.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

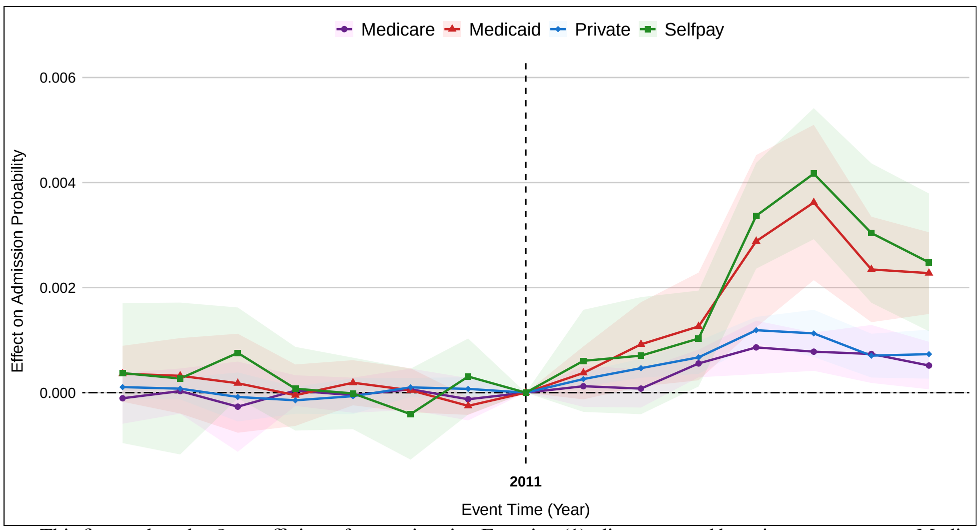
Table 3 and Figure 5 present estimates disaggregated by primary payer type. The largest absolute increases occurred among Medicaid and self-pay patients. For Medicaid patients, the supply shock increased heroin overdose admissions by 0.20 percentage points (SE = 0.0005), or 83 percent above the 0.24 baseline mean. Self-pay patients saw an increase of 0.18 percentage points (SE = 0.0002), representing a 69 percent increase above the 0.26 percent baseline. Medicare and privately insured patients experienced smaller effects of 0.06 and 0.07 percentage points, respectively.

The event-study estimates in Figure 5 reveal parallel pre-trends for all payer groups, followed by a pronounced and sustained rise for Medicaid and self-pay patients, peaking at roughly 0.4 percentage points above baseline by 2016–2017. The concentration of effects among these groups reflects both higher underlying exposure risk and greater barriers to treatment and harm reduction (Bremer et al., 2023; Bowser et al., 2024).

From a fiscal perspective, Medicaid costs fall directly on the public sector, while self-pay admissions often lead to uncompensated care and medical debt that are indirectly absorbed by public funds through subsidies and cost shifting (Pryor et al., 2003; Anderson, 2007; Himmelstein et al., 2009). Consequently, the two groups most adversely affected by the heroin supply shock also generate substantial downstream fiscal costs.

Beginning in 2014, many patients who would previously have been classified as self-pay gained Medicaid coverage under the Affordable Care Act's Medicaid expansion (Cole et al., 2018; Hamer et al., 2022). The expansion is unlikely to confound our results because

Figure 5. Event Study by Insurance Type: Heroin Overdose Admissions



Notes: This figure plots the β_τ coefficients from estimating Equation (1), disaggregated by primary payer type: Medicare, Medicaid, private insurance, and self-pay. Each line shows the differential change in heroin-related inpatient admissions between treatment and control regions relative to the reference period (2011), which immediately precedes the supply shock. Event time spans from 2004 to 2019, excluding 2015 due to data unavailability. All specifications include Census division and calendar-year fixed effects, as well as individual-level demographic controls and discharge weights.

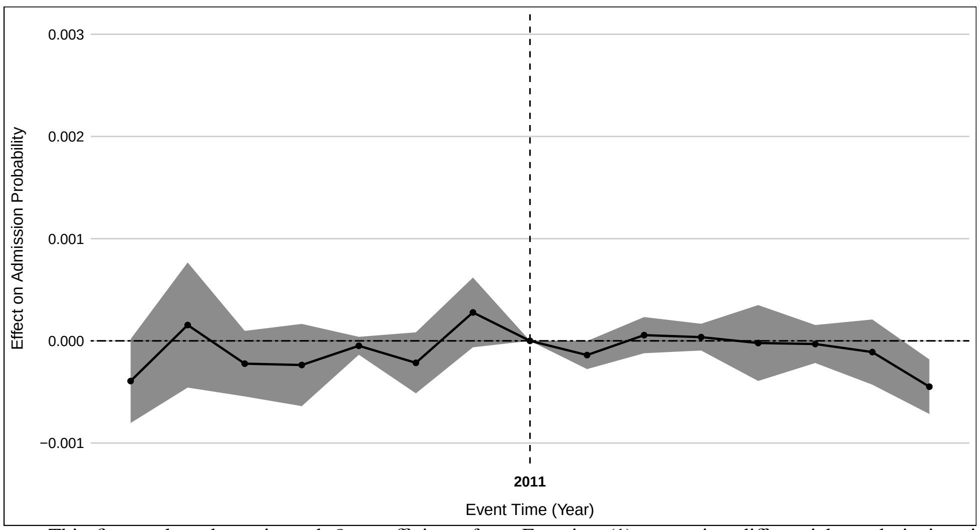
it did not differentially affect Eastern versus Western regions.[17] However, it likely shifted some of the financial burden from hospitals absorbing uncompensated care to the public sector through Medicaid financing.

V. Discussion

A. Placebo Tests

Alcohol Poisoning

Figure 6. Event Study: Alcohol Poisoning Overdose Admissions



Notes: This figure plots the estimated β_τ coefficients from Equation (1), capturing differential trends in inpatient admissions for alcohol poisoning between treatment regions (east of the Mississippi River) and control regions (west), relative to the omitted reference period (2011), which immediately precedes the supply shock. The event time spans from 2004 to 2019, excluding 2015 due to data unavailability. All specifications include Census division and calendar year fixed effects, as well as individual-level demographic controls and discharge weights.

As a falsification exercise, Figure 6 examines alcohol poisoning admissions. Alcohol use correlates with illicit drug consumption through shared confounders such as mental health and socioeconomic stressors (Jane-Llopis and Matytsina, 2006; Conway et al., 2016), but alcohol is sold through a legally regulated supply chain that should not be affected by heroin market disruptions. The estimated coefficients are small, statistically insignificant, and fluctuate around zero in both the pre- and post-shock periods. No structural break emerges after 2012, and the magnitudes are an order of magnitude smaller than those for heroin. This null result strengthens our identification by showing

[17] Our data do not permit state-level identification, but Donahoe and Soliman (2025) confirm that ACA expansion is unlikely to confound outcomes attributed to the heroin supply shock.

that the empirical framework does not mechanically generate spurious treatment–control divergence in unrelated markets.

Cocaine

Cocaine-related overdoses provide a further placebo test. Because specific ICD codes for cocaine overdoses are consistently available only after the 4th quarter of 2010, long pre-trends with annual data cannot be fully assessed. Nonetheless, post-2012 estimates show no systematic increase in cocaine overdose inpatient admissions when using quarterly data.[18] This absence of spillover is consistent with evidence of drug-specific persistence: most users maintain stable consumption patterns, and large shocks in one market rarely induce major substitution into another, particularly when the substances have different pharmacological effects. Some overlap may arise from "speedballing", the combined use of heroin (a depressant) and cocaine (a stimulant), which some users mistakenly believe offsets adverse effects (Leri et al., 2003). The lack of measurable increases in cocaine admissions, therefore, reinforces the interpretation that the post-2012 rise in overdose hospitalizations was driven primarily by heroin quality shocks rather than expanded overall drug use.

B. *Spillover Effects*

Fentanyl

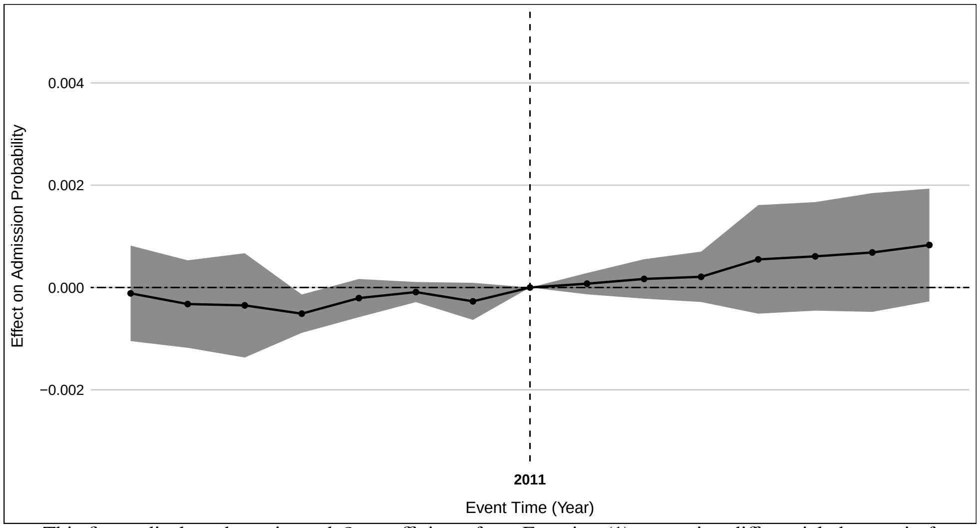
Figure 7 shows suggestive but statistically insignificant post-shock changes in fentanyl-related inpatient admissions, despite contemporaneous evidence of rising fentanyl adulteration in Eastern heroin markets after the quality shock. Historical accounts indicate that most established heroin users actively avoided fentanyl due to its distinct pharmacological profile and heightened overdose risk, making large-scale voluntary uptake improbable. Spillovers were therefore more plausibly the result of involuntary exposure through adulterated heroin rather than deliberate substitution.

Measurement constraints likely further attenuate estimable effects. In the NIS, overdoses involving both heroin and fentanyl are often coded solely as heroin-related when heroin is deemed the primary clinical finding (Slavova et al., 2014; Rudd et al., 2016). This practice embeds many fentanyl-involved cases within heroin-coded admissions, producing undercounts in fentanyl-specific categories and biasing estimated effects toward zero. Consequently, the absence of a statistically significant effect should not be interpreted as evidence against a meaningful rise in fentanyl-involved overdoses in the Eastern U.S.

In addition, fentanyl adulteration could increase the frequency of heroin use. Although both substances are potent opioids, fentanyl acts far more rapidly and wears off more quickly than heroin. Clinical pharmacokinetic studies show that fentanyl's onset of action can occur within several minutes, with an effective duration of roughly one hour (Mather, 1983; Varvel et al., 1989; Panagiotou and Mystakidou, 2010). In contrast,

[18] See Figure A4 in Appendix.

Figure 7. Event Study: Fentanyl Overdose Admissions



Notes: This figure displays the estimated β_τ coefficients from Equation (1), capturing differential changes in fentanyl-related inpatient admissions between treatment regions (east of the Mississippi River) and control regions (west), relative to the omitted reference period (2011), which immediately precedes the supply shock. The event time spans from 2004 to 2019, excluding 2015 due to data unavailability. All models include Census division and calendar year fixed effects, as well as individual-level demographic controls and discharge weights.

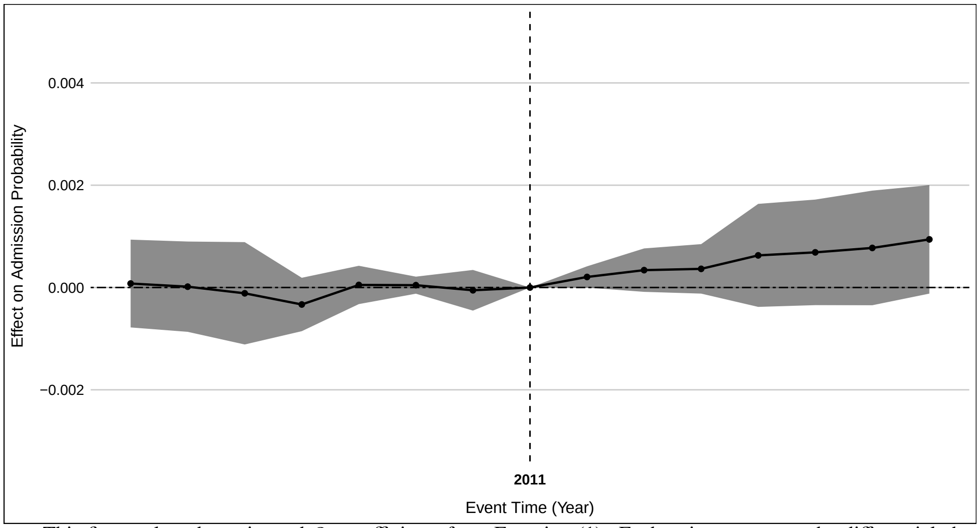
heroin is metabolized more slowly and produces a subjective "high" lasting several hours (Rook et al., 2006a,b). When heroin is adulterated with fentanyl, the rapid offset of fentanyl's effect may induce users to re-dose more frequently to maintain euphoria or avoid withdrawal. However, these pharmacokinetic estimates derive from controlled medical settings rather than illicit-use contexts, and they describe biochemical duration rather than observed dosing behavior. Real-world illicit conditions, where purity and dosage vary widely, are likely even more hazardous.

A growing body of evidence highlights fentanyl's extreme potency, minimal production costs, and independence from natural opium inputs, all of which make it more scalable and lethal than heroin (DEA, 2015a; Ciccarone et al., 2017; Pardo and Reuter, 2020; CDC, 2024; Donahoe and Soliman, 2025). Although our estimates show no statistically significant effects, its role in the 2012 heroin supply shock was likely latent rather than absent. As fentanyl has since become the leading driver of the US opioid epidemic, even localized supply disruptions can rapidly cascade into nationwide public-health crises—much as the heroin shock documented here did, and could with even greater intensity.

Methadone

Controlled prescription drugs (CPDs) have long been ranked among the most significant drug threats in the U.S. (DEA, 2015a). Many prescription opioid abusers who transitioned to heroin cited heroin's similar high, lower price, and greater accessibility. Others

Figure 8. Event Study: Methadone Overdose Admissions



Notes: This figure plots the estimated β_τ coefficients from Equation (1). Each point represents the differential change in methadone-related inpatient admission rates between treatment regions (east of the Mississippi River) and control regions (west), relative to the omitted reference period (2011), which immediately precedes the supply shock. The event time spans from 2004 to 2019, excluding 2015 due to data unavailability. All specifications include Census division and calendar year fixed effects, as well as individual-level demographic controls and discharge weights.

initiated heroin use directly, bypassing prescription misuse but contributing to rising heroin demand. Market changes in CPDs, such as the 2010 OxyContin reformulation, can therefore potentially influence or be influenced by illicit opioid markets.[19]

In the NIS, methadone[20] is the only CPD outcome we can directly observe. Figure 8 shows flat pre-treatment trends with no statistically significant East–West differences before 2012. After the heroin supply shock, there is a mild upward drift in post-period means, but the estimates remain statistically insignificant. Because methadone is distributed through a legal, tightly regulated treatment system, its inpatient admission profile is insulated from illicit supply shocks. Small post-shock increases may reflect greater engagement with opioid use disorder treatment, either in response to increased heroin harms or due to policy and clinical outreach in affected regions.

C. *Implications*

The absence of any measurable change in alcohol poisoning and cocaine overdose admissions strengthens the credibility of our identification strategy. This placebo result mirrors the null or near-null findings for other drug markets that either operate under legal regulation (methadone) or are subject to measurement attenuation and persistent

[19]Some CPD abusers turn to heroin as a cheaper alternative when prescription opioids become more expensive or restricted (DEA, 2015a).

[20]Methadone is a synthetic opioid used medically to treat chronic pain and opioid use disorder.

preferences (fentanyl). Moreover, prior evidence indicates that naloxone[21] rollout, a potentially confounding public health intervention, did not disproportionately change between Eastern and Western states over this period (Guy, 2019). Taken together, these results reinforce the conclusion that the morbidity increase documented in the main results is specific to heroin and unlikely to reflect broader shifts in substance use, treatment access, or healthcare utilization.

D. Conceptual Framework: Market Failures and Supply Shocks

Our results highlight three economic mechanisms through which illicit drug markets transmit supply shocks into social costs: underprovision of public goods, asymmetric information, and negative externalities.

Public Goods and Surveillance.

Drug surveillance is a classic public good: it generates large positive spillovers by enabling timely intervention and reducing downstream health costs, yet is chronically underprovided because its benefits are diffuse. The dismantling of federal programs such as the Arrestee Drug Abuse Monitoring Program (ADAM) and the HDMP illustrates this failure. Rising admissions after 2016 coincided with the termination of many such monitoring systems, consistent with the hypothesis that reduced surveillance exacerbated vulnerability to adulterated supply. While we cannot claim a direct causal link, the contrast in magnitudes is stark: these programs operated on annual budgets of only a few million dollars,[22] whereas the welfare losses we estimate in section VI from the heroin supply shock exceed billions. Chronic underinvestment in surveillance thus leaves societies exposed to hidden but extremely costly shocks in illicit supply chains.

Asymmetric Information and Adulteration.

The heroin–fentanyl dynamic illustrates a textbook “lemons” problem. Drug users cannot observe product quality or adulteration at the point of purchase, learning only after consumption—often at lethal cost. This asymmetric information produces involuntary fentanyl exposure, inflated morbidity/mortality, and inefficient market outcomes. In standard models, adverse selection drives high-quality products out of the market; in illicit drug markets, users cannot exit, so unobservable quality manifests instead as excess hospital admissions. Our evidence from this illicit drug market shift shows how quickly such information failures can translate into measurable public health and economic costs.

[21] Naloxone is a fast-acting opioid overdose reversal medication.

[22] For example, the ADAM program cost $6–7 million annually in the late 1990s, with ADAM II budgets of about $10 million in 2011–2012 (Office of National Drug Control Policy , ONDCP,O; U.S. Department of Justice, 2011). The DEA’s HDMP and Heroin Signature Program together cost roughly $1.3 million annually in the early 2000s, including laboratory analyses (U.S. Government Accountability Office , GAO).

Externalities and Community Spillovers.

Finally, the heroin supply shock generated negative externalities extending far beyond individual users. In the next section, we document fiscal burdens through Medicaid, uncompensated hospital care, and disability programs. But the indirect costs are broader still: reduced property values and disinvestment in affected neighborhoods, higher policing and incarceration expenditures, intergenerational deficits in children's outcomes, and weaker local economic dynamism. These costs, borne collectively by households and local governments, exemplify Pigouvian externalities. They underscore that opioid crises are not merely individual tragedies but systemic shocks with far-reaching community consequences.

These mechanisms demonstrate that our empirical case is not unique. It reflects general principles of public goods underprovision, asymmetric information, and externalities that characterize illicit markets globally. When institutions fail to monitor and contain supply shocks, localized disruptions can rapidly escalate into large-scale social costs.

E. One Ambiguous Mechanism: Price

One potential channel contributing to the rise in hospital admissions is price. In theory, a decline in heroin prices, particularly given widespread fentanyl adulteration, which reduced production costs, could have lowered the cost of entry and expanded the user base. However, DEA intelligence suggests a more dangerous dynamic. As traffickers increasingly mixed fentanyl into local heroin supplies, they often charged higher prices because the adulterated heroin delivered a much more intense high. This allowed suppliers to offer stronger potency at lower cost and exposed users to significantly greater overdose risk (DEA, 2019).

Empirical validation is limited by data constraints. The most credible and publicly available heroin price data come from the HDMP. As shown in Table A2, the reported price per pure gram of heroin, sourced from both Mexican and Colombian suppliers, remained relatively stable between 2003 and 2016 across the major cities. This observed stability suggests that price effects, while theoretically possible, were unlikely to be the primary driver of the post-shock rise in hospitalizations. Although this evidence is consistent with our placebo tests of the cocaine market, which also point to stability in overall consumption rather than a surge in quantity driven by lower prices, definitive conclusions are difficult to draw for several reasons.

First, the HDMP typically involved only a small number of undercover street-level purchases per city, raising concerns about representativeness. Second, HDMP reporting is standardized to the price per pure gram of heroin rather than the retail price actually paid by users, limiting our ability to capture real-world purchasing conditions or assess user-level price sensitivity. Third, the HDMP was discontinued after 2016, preventing analysis of long-run price dynamics as the heroin market shock fully unfolded.

These constraints again highlight the importance of maintaining robust surveillance programs like the HDMP and NDTA. Continued collection of timely, granular, and representative data on drug prices and purity is essential for quantifying the mechanisms driving changes in illicit drug markets and for designing more targeted interventions in response to evolving public health threats.

F. Global Relevance

While our analysis focuses on the U.S., the mechanisms we identify extend well beyond the opioid crisis. Illicit drug markets worldwide share three structural features: fragmented supply chains, limited quality control, and weak surveillance. These conditions make them similarly vulnerable to sudden supply-side shocks and adulteration.

Across East and Southeast Asia, methamphetamine markets have expanded rapidly over the past two decades. Production is decentralized, precursor chemicals are trafficked across porous borders, and enforcement pressures often shift rather than eliminate supply. Users face limited information about purity or adulterants, leaving markets vulnerable to abrupt changes in potency or price that can generate large and uneven health burdens.

In West Africa and the Sahel, tramadol has emerged as a dominant synthetic opioid, trafficked in bulk through informal and poorly regulated channels. Weak pharmaceutical oversight and widespread counterfeit production have created an environment where adulteration and mislabeling are pervasive. As in the U.S. heroin case, the lack of reliable surveillance systems leaves governments unable to detect changes in market composition until morbidity and mortality rates are already rising.

These parallels suggest that the U.S. illicit drug market shift is not an isolated event but a cautionary tale. In settings where synthetic drug markets are growing rapidly and state surveillance is limited, the combination of adulteration, asymmetric information, and public goods underprovision can yield sudden and catastrophic social costs. The broader implication is that drug supply shocks are not unique to opioids or to the U.S.; they are an inherent risk in any illicit market where weak institutions allow volatility in product quality to propagate unchecked.

VI. Welfare Implications

To quantify the economic burden of the heroin supply shock, we distinguish between *direct* medical costs arising from overdose incidents and *indirect* costs operating through mortality, labor markets, families, and communities. Table 4 summarizes our estimates.

Direct Costs

Our event-study estimates imply that heroin-related inpatient admissions rose by approximately 0.11 percentage points in affected regions (i.e., east of the Mississippi River) between 2012 and 2019. Given that the National Inpatient Sample records roughly 35 million hospital discharges annually, about 70% of which occur in the Eastern U.S. (Healthcare Cost and Utilization Project, 2024), this corresponds to roughly 215,000 excess admissions over the eight-year period. At an average cost of $12,000 per heroin overdose inpatient admission (Premier, 2019),[23] the total inpatient burden exceeds $2.5

[23]The mean of the total medical bill per heroin overdose is around $40,000 in the NIS, but this reflects hospital charges (list prices billed). Economic costs, which approximate actual resource use and payments, are substantially lower. We

billion.

A large share of overdose encounters occur in the emergency department (ED). Within our sample, we focus on non-elective admissions—patients who present to the ED and are subsequently admitted for inpatient care. In addition to these admissions, however, many overdose patients are treated and released from the ED without being admitted. Following prior estimates, we assume that for every overdose admission, there are approximately 0.75 such non-admitted ED visits (Weiss et al., 2017; Vivolo-Kantor et al., 2020). With an average cost of just over $1,500 per visit (including ambulance and medical services) (Guy Jr et al., 2018), these cases add several hundred million dollars in expenditures. Clinical evidence further suggests that roughly 4% of nonfatal overdoses result in permanent disability (Adams et al., 2020). Using lifetime cost estimates from the brain-injury literature ($115,300 per disabled survivor; Lemsky and ABPP-CN, 2021), this adds nearly $1 billion in long-term costs.

Other direct medical costs further increase the total burden. Many survivors require detoxification or ongoing medication-assisted treatment with methadone or buprenorphine. With average program costs of $6,000–$8,000 annually (Florence et al., 2016), the cumulative fiscal impact for the excess overdose cohort likely amounts to hundreds of millions of dollars. Neonatal abstinence syndrome (NAS) cases, which potentially rose in the East during this period, cost $25,000–$35,000 per hospitalization (Corr and Hollenbeak, 2017), also contributed substantially to the overall medical expenditures.

Summing across these categories yields a conservative estimate of $4 billion in direct health-care costs. This estimate likely understates the true direct burden because many fentanyl-adulterated overdoses were misclassified due to ICD coding limitations, some overdoses did not result in hospitalization, and spillovers into other illicit drug markets were not captured. Moreover, Figure 3 shows that elevated admissions persisted through 2019. Without major intervention, ongoing inpatient costs alone could exceed $0.5 billion annually.[24]

Indirect Costs

Indirect costs are even larger and more diffuse than the direct medical expenditures. The most immediate channel is mortality. Applying a 5%[25] fatality rate to the excess overdose population implies nearly 10,000 additional deaths.[26] Using the central federal estimate of the Value of a Statistical Life (about $10 million per life) (Kniesner and Viscusi, 2019; U.S. Department of Transportation, 2024; U.S. Environmental Protection Agency, 2024), this translates into roughly $50–100 billion in long-run social losses from mortality alone.

A second channel involves reduced labor market activity among survivors. Opioid

therefore use an average cost of about $12,000 per inpatient admission, consistent with national estimates from HCUP and Premier data.

[24]The elevated effect we identify is unlikely to have ended in 2019; our sample stops then only because the onset of the COVID-19 pandemic introduces confounding factors that are difficult to separate from the underlying trend.

[25]We observe a 4.95% death rate among heroin overdose inpatients from the NIS, which is consistent with prior studies (Florence et al., 2016; Stevens et al., 2017; Lewer et al., 2021; Casillas et al., 2024).

[26]This estimate also aligns closely with Donahoe and Soliman (2025).

Table 4— Estimated Welfare Losses from the Heroin Supply Shock (2012–2019)

Category	**Unit / Assumption**	**Magnitude**
Panel A. Direct Medical Costs		
Inpatient admissions (215,000 cases)	\$12,000 per admission	\$2.5 billion
Non-admitted ED visits (0.75 per admission)	\$1,500 per visit	\$0.2–0.3 billion
Permanent disability (4% of admissions)	\$115,300 per case	\$1.0 billion
Medication-assisted treatment (MAT)	\$6,000–\$8,000 per patient-year	\$0.3–0.5 billion
Neonatal Abstinence Syndrome (NAS)	\$25,000–\$35,000 per case	\$0.1–0.2 billion
Subtotal: Direct Costs		**Approximately \$4 billion**
Panel B. Indirect and Social Costs (Unmonetized)		
Mortality (5% fatality rate)	≈9,000 additional deaths	Not monetized
Productivity losses (non-fatal survivors)	Lower labor force participation, SSDI uptake	Not monetized[a]
Intergenerational spillovers	Foster care, reduced child education and health	Not monetized
Community externalities	Crime, housing prices, comorbidities, cohesion	Not monetized
Subtotal: Indirect Costs		**Presumably much larger than direct costs**
Total Welfare Loss (conservative)		**At least \$4 billion**

Notes: Direct costs include excess inpatient and emergency visits, long-term disability, MAT programs, and NAS care (Premier, 2019; Guy Jr et al., 2018; Adams et al., 2020; Lemsky and ABPP-CN, 2021; Florence et al., 2016; Corr and Hollenbeak, 2017). Indirect costs encompass excess mortality (about 9,000 additional deaths), reduced labor market activity and increased reliance on social insurance (Krueger, 2017; Harris et al., 2020), intergenerational spillovers through foster care and reduced child human capital (Quast et al., 2019; Meinhofer et al., 2024), and community-level harms such as crime, housing disinvestment, comorbidities, and weakened social cohesion (Krebs et al., 2017; Ropero-Miller and Speaker, 2019; Chen et al., 2022; Savinkina et al., 2023; D'Lima and Thibodeau, 2023; Schranz et al., 2018; Zibbell et al., 2018; Corr and Hollenbeak, 2017; Goplerud et al., 2017; Christie, 2021). Because these broader costs are not monetized, the totals reported in the table should be viewed as lower bounds. Given the scale of excess mortality and spillovers, overall welfare losses plausibly exceed \$10 billion, and may be several times larger, depending on how fatalities are valued.

[a] For illustration, a 20% decline in productivity ((Florence et al., 2016)) and assumed average annual earnings of \$50,000 over 6–8 years would imply near 10 billion dollars in lost earnings, but we do not treat this as a central estimate.

misuse lowers labor force participation among prime-age workers[27] and increases reliance on Social Security Disability Insurance and other public programs (Krueger, 2017; Harris et al., 2020). The excess overdoses we identify likely exacerbated these patterns in the Eastern U.S., contributing to regional employment declines and productive losses beyond the national pattern.

Intergenerational spillovers represent another important dimension of social cost. Parental overdose raises foster care placements and child protective services involvement, which is associated with long-term deficits in children's education and health (Quast et al., 2019; Meinhofer et al., 2024). These mechanisms reduce human capital accumulation in the next generation, transmitting the burden of the shock across cohorts.

Finally, community-level externalities further amplify the harm. Elevated overdose rates are linked to higher crime, incarceration, and policing costs, as well as diminished neighborhood safety (Krebs et al., 2017; Ropero-Miller and Speaker, 2019; Chen et al., 2022; Savinkina et al., 2023). Opioid mortality depresses housing prices and discourages investment (D'Lima and Thibodeau, 2023), eroding local wealth and tax bases. Health systems face additional burdens from comorbid conditions such as HIV (Schranz et al., 2018) and hepatitis C (Zibbell et al., 2018), much of which falls on Medicaid and other public insurance programs. Persistent misuse also undermines local economic dynamism by raising turnover and absenteeism (Goplerud et al., 2017), discouraging in-migration, and weakening social cohesion (Christie, 2021).

Although we provide some figures for indirect costs, we do not treat them as central estimates but rather as benchmarks for the potential scale. These channels highlight that the indirect costs of the heroin supply shock dwarf even the already substantial direct medical expenditures. They encompass excess mortality, lost productivity, intergenerational human capital deficits, fiscal strain on social insurance systems, and the erosion of community stability. The true social burden is plausibly an order of magnitude larger than what can be measured through baseline health-care costs alone.

VII. Conclusion

This paper documents the substantial public health and fiscal consequences of a major supply-side shock in the U.S. illicit drug market. In 2012, Mexican DTOs displaced Colombian suppliers and assumed control of white powder heroin distribution in the Eastern U.S. This transition introduced greater variability in heroin purity and facilitated widespread fentanyl adulteration. Using discharge-level data from the National Inpatient Sample and an event-study framework, we find that heroin-related inpatient admissions rose by 0.11 percentage points in the East following the supply shock. The effects were concentrated among Medicaid and self-pay patients, underscoring significant public sector exposure and financial strain, and were largest among White populations whose prior exposure to prescription opioids left them especially vulnerable to transition into illicit markets.

[27] For illustration, if survivors experienced even a 20% productivity decline sustained over eight years (Florence et al., 2016), implied earnings losses would run into the billions.

Placebo tests using alcohol- and cocaine- related admissions reveal no comparable effects, reinforcing the causal interpretation. We estimate direct medical costs of roughly $4 billion, primarily from excess inpatient and emergency care, disability, and treatment needs. Beyond these measurable expenditures, the shock could impose far larger unmonetized social costs: thousands of excess deaths, diminished labor force participation, intergenerational human capital losses, rising fiscal pressures on social insurance, depressed housing markets, higher crime and policing expenditures, and weakened community stability. Overall, these results indicate that the welfare burden of the heroin market shock cannot be captured by medical expenditures alone, as the unmeasured social and economic spillovers are likely several times larger.

Anticipation, rather than reaction, must guide opioid policy. Reactive interventions are insufficient to contain the fallout of sudden market disruptions. Preventing the next wave of synthetic opioid harm requires proactive investments in early-warning surveillance systems and real-time dissemination of drug composition data. Yet many such programs, including the Arrestee Drug Abuse Monitoring Program and the Heroin Domestic Monitoring Program, were dismantled as the crisis unfolded, leaving policymakers blind to emerging threats (Carnevale Associates, 2025). Rebuilding and sustaining these monitoring systems is essential. At the same time, harm-reduction infrastructure, including naloxone distribution, fentanyl test strips, and safe consumption sites, must be expanded and equitably deployed. Durable policy responses must also address the broader socioeconomic consequences of opioid use, from disability care to workforce reintegration and family support.

More broadly, our findings highlight that illicit drug market shocks are not merely criminal justice issues but core public health and economic challenges. The 2012 heroin supply shock illustrates the high costs of delayed response: when institutions underinvest in surveillance and harm reduction, volatility in illicit supply chains can metastasize into large-scale social costs. Proactive investment in public goods is therefore far more cost-effective than reactive crisis management.

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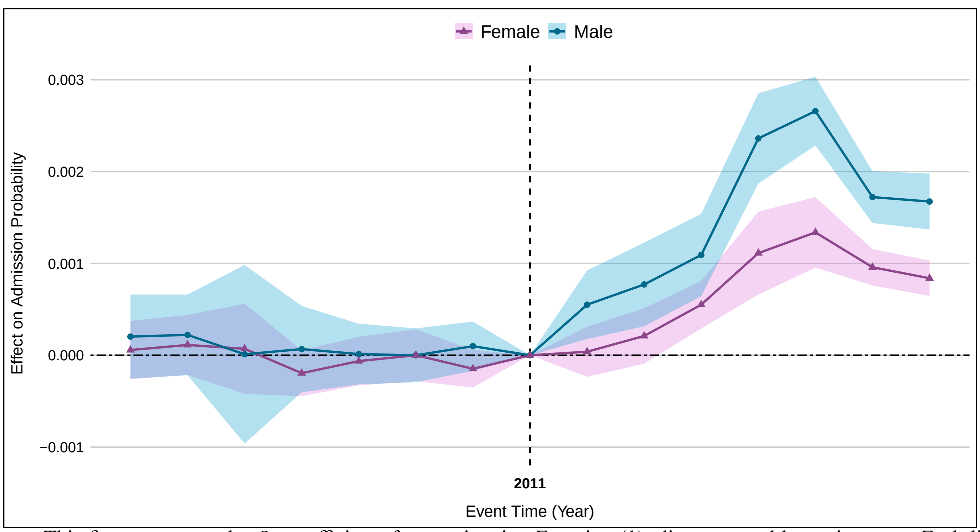
Appendix

Additional Heterogeneity Analyses

The aggregate effects documented in the main text mask important variation across subgroups defined by race and insurance status. These differences reflect how illicit market shocks interact with pre-existing disparities in health care, socioeconomic status, and access to treatment. We extend the analysis along two additional axes: sex and urban–rural residence. Both dimensions are motivated by established epidemiological patterns and market dynamics. Men have historically reported higher rates of heroin use and overdose than women, while urban and rural areas differ sharply in the organization of supply chains and the availability of treatment resources (Cicero et al., 2014; Mars et al., 2016). By probing these subgroups, we assess whether the supply shock redistributed harms in ways that deepen existing disparities and impose uneven fiscal burdens.

A1. Heterogeneity by Sex

Figure A1. Event Study by Sex: Heroin Overdose Admissions



Notes: This figure presents the β_τ coefficients from estimating Equation (1), disaggregated by patient sex. Each line shows the estimated difference in the probability of inpatient admission for heroin-related overdose between treatment regions (east of the Mississippi River) and control regions (west), relative to the omitted reference period (2011), which immediately precedes the supply shock. The event time spans from 2004 to 2019, excluding 2015 due to data unavailability. The model includes Census division and calendar year fixed effects, as well as individual-level demographic controls and discharge weights.

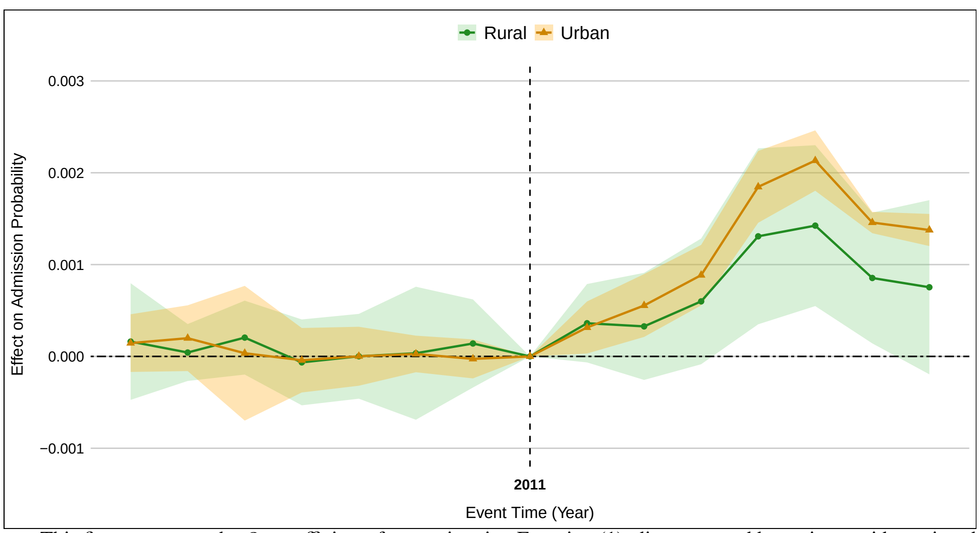
Appendix Figure A1 plots event-time coefficients separately for male and female patients. Pre-treatment coefficients are statistically indistinguishable from zero for both groups, supporting parallel trends. Following the 2012 heroin market shock, both men and women in the East exhibit significant increases in heroin-related inpatient admissions relative to their Western counterparts, though the magnitude is larger for men.

This gender gap is consistent with national evidence that men historically report higher heroin use and overdose prevalence (Cicero et al., 2014). Men are also more likely to engage in high-frequency injection behaviors, which heighten sensitivity to sudden changes in heroin purity. While women experienced smaller absolute increases, they nonetheless faced meaningful relative rises in overdose risk. These patterns suggest that supply shocks can exacerbate existing gender disparities in opioid-related morbidity, potentially due to differences in risk behaviors, treatment engagement, and access to harm-reduction resources.

A2. Heterogeneity by Urban–Rural Status

Appendix Figure A2 compares event-time estimates for patients residing in urban versus rural counties, classified using USDA Rural–Urban Continuum Codes. Prior to 2012, both groups displayed flat and indistinguishable pre-trends. After the heroin market shock, urban areas in the East exhibit sharp and immediate increases in heroin-related inpatient admissions, while rural areas experience smaller and more gradual increases.

Figure A2. Event Study by Urban/Rural Status: Heroin Overdose Admissions



Notes: This figure presents the β_τ coefficients from estimating Equation (1), disaggregated by patient residence in urban versus rural areas. Each line shows the estimated difference in the probability of inpatient admission for heroin-related overdose between treatment regions (east of the Mississippi River) and control regions (west), relative to the omitted reference period (2011), which immediately precedes the supply shock. The event time spans from 2004 to 2019, excluding 2015 due to data unavailability. The model includes Census division and calendar year fixed effects, as well as individual-level demographic controls and discharge weights.

These dynamics are consistent with the spatial organization of drug supply chains. Urban markets, characterized by higher turnover and shorter distribution chains, are typically the first to absorb shifts in supply quality. Rural areas, by contrast, may be initially insulated due to reliance on slower, localized networks, but eventually experience similar risks as adulterated products diffuse outward (Mars et al., 2016). The modest, delayed but persistent rise in rural admissions underscores that large-scale shocks are not confined to metropolitan centers.

While urban systems absorbed the initial volume surge, rural hospitals, often operating with limited resources and thin margins, faced intensified fiscal stress from uncompensated care. This uneven burden highlights a broader distributional consequence: urban centers bear the brunt of initial morbidity, whereas rural communities ultimately face long-run fragility. These findings reinforce the need for geographically sensitive harm-reduction and surveillance strategies, which we return to in the policy discussion.

List of Abbreviations

Abbreviation	Meaning
ADAM	Arrestee Drug Abuse Monitoring Program
ACA	Affordable Care Act
AHRQ	Agency for Healthcare Research and Quality
BTH	Black Tar Heroin
CDC	Centers for Disease Control and Prevention
CPD	Controlled Prescription Drugs
DEA	Drug Enforcement Administration
DiD	Difference-in-Differences
DTO	Drug Trafficking Organization
ED	Emergency Department
EPA	Environmental Protection Agency
HDMP	Heroin Domestic Monitor Program
HSP	Heroin Signature Program
ICD	International Classification of Diseases
MAT	Medication-Assisted Treatment
NAS	Neonatal Abstinence Syndrome
NIS	National Inpatient Sample
ONDCP	Office of National Drug Control Policy
VSL	Value of a Statistical Life
WPH	White Powder Heroin

Table A1— Summary Statistics by Treatment Status

	Treatment	Control
Demographics		
Age	46.66	47.02
Female	0.491	0.487
Urban	0.854	0.843
Race/Ethnicity		
White	0.584	0.625
Black	0.140	0.235
Hispanic	0.200	0.087
Asian or Pacific Islander	0.035	0.012
Native American	0.014	0.004
Other	0.027	0.036
Payer Type		
Medicare	0.181	0.211
Medicaid	0.196	0.208
Private insurance	0.400	0.412
Self-pay	0.131	0.112
No charge	0.012	0.014
Other	0.081	0.043
Overdose Diagnoses		
Heroin	0.0010	0.0010
Fentanyl	0.0030	0.0023
Prescription opioid	0.0019	0.0016
Methadone	0.0013	0.0011
Cocaine	0.0007	0.0008
Alcohol poisoning	0.0019	0.0016
Observations	5,187,430	11,102,217

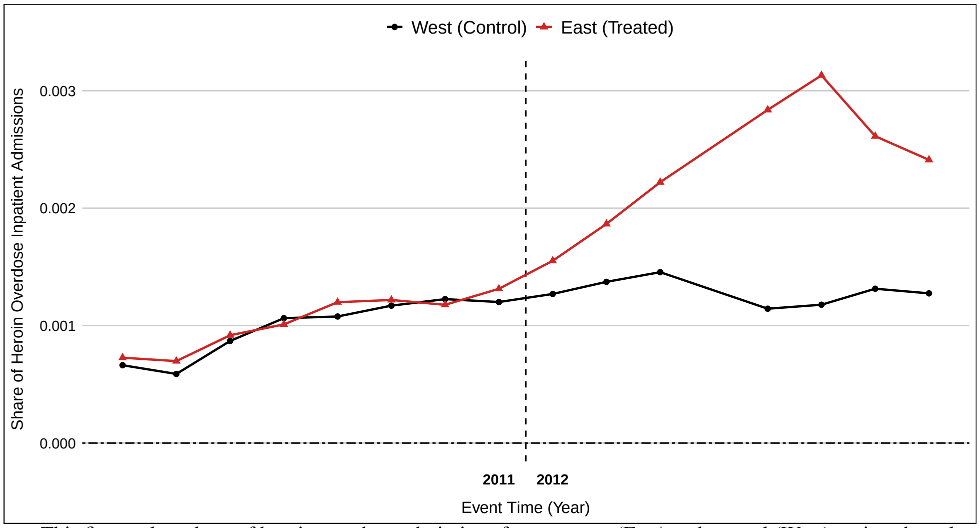
Notes: The table reports mean values for the treatment group (regions east of the Mississippi River) and the control group (regions west of the Mississippi River) using patient-level records from the National Inpatient Sample (NIS) for 2004–2019, excluding 2015. "Urban" is an indicator for residence in a metropolitan statistical area. Race/ethnicity categories are mutually exclusive. Income quartiles refer to the median household (HH) income of the patient's ZIP code, as provided by the NIS. Payer type is based on the primary expected source of payment at admission. Overdose diagnoses are binary indicators based on ICD-9/ICD-10 codes for the corresponding substance. Sample sizes reflect discharge-level observations and are unweighted.

Table A2— Heroin Exhibits, Purity, and Price per mg by Origin and Year, 2003–2016.

	Year													
Measure	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
Mexican Origin														
Exhibits	317	342	278	290	327	351	322	309	296	339	357	287	600	543
Purity (%)	26.3	27.9	32.5	30.0	33.1	26.8	24.7	14.7	16.8	17.6	20.3	21.1	29.0	31.5
Price ($/mg)	0.75	0.97	0.66	0.77	0.81	1.06	1.11	2.00	1.35	1.40	1.13	1.15	0.93	0.84
South American Origin														
Exhibits	470	420	396	418	438	403	341	346	323	375	334	34	42	14
Purity (%)	41.8	32.5	37.3	36.1	35.0	33.6	33.6	25.9	31.1	35.3	35.1	31.1	39.1	25.8
Price ($/mg)	0.77	1.00	0.81	1.04	1.00	1.07	1.28	1.75	1.18	1.15	1.04	1.03	1.05	1.09
Inconclusive Origin														
Exhibits	–	–	–	–	–	–	–	–	–	–	–	303	178	106
Purity (%)	–	–	–	–	–	–	–	–	–	–	–	38.0	27.5	20.1
Price ($/mg)	–	–	–	–	–	–	–	–	–	–	–	1.07	1.19	1.69

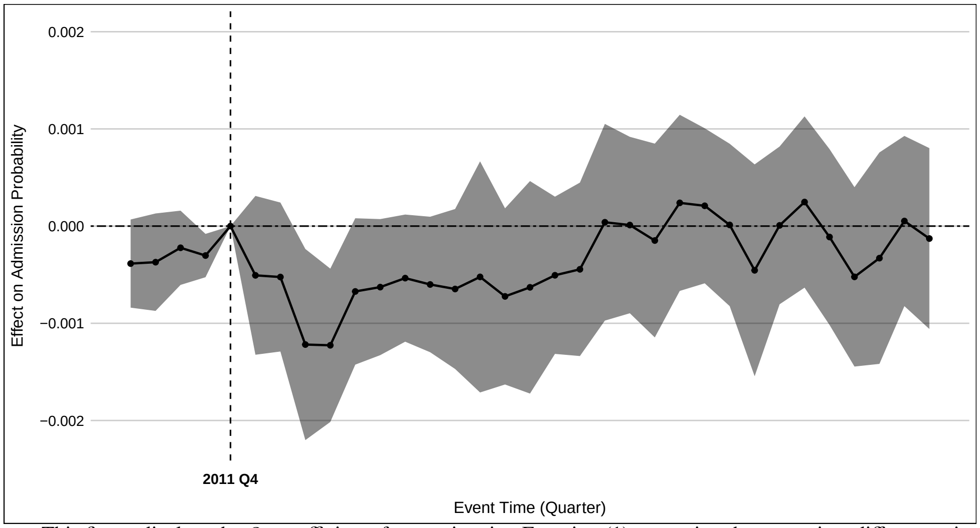
Notes: Data are drawn from the DEA's Heroin Domestic Monitor Program, which conducts undercover retail-level purchases of heroin in major U.S. cities and reports exhibit counts, average purity, and standardized price per milligram of pure heroin. Reported prices are adjusted for purity and do not reflect the retail street price actually paid by users. Because the HDMP typically involves a small number of purchases per site, estimates may not be fully representative of local markets. Reporting for South American origin heroin declines sharply after 2013, reflecting the near-complete displacement of Colombian suppliers by Mexican traffickers. "Inconclusive origin" denotes exhibits that could not be chemically attributed to a specific source region.

Figure A3. Total Inpatient Admissions by Region



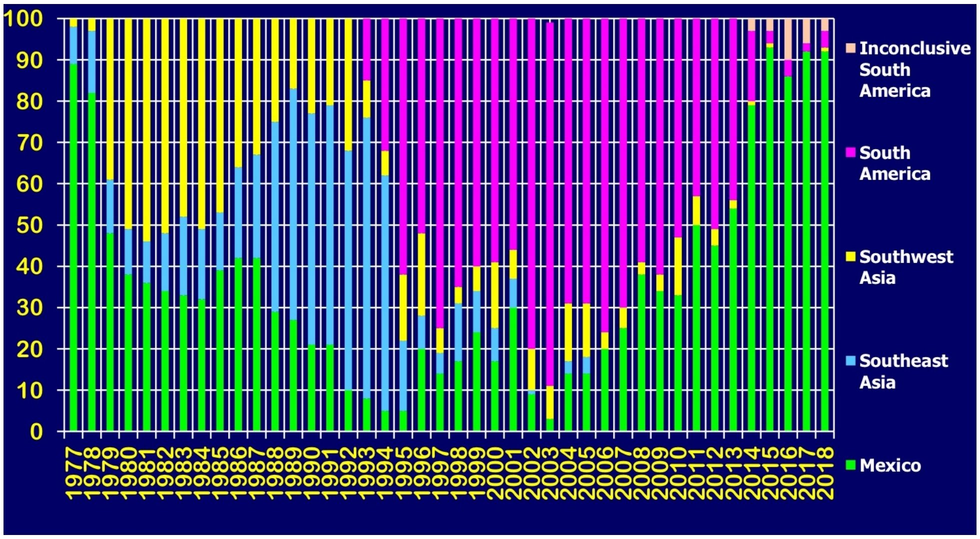
Notes: This figure plots share of heroin overdose admissions for treatment (East) and control (West) regions by calendar year. Points and lines denote annual series by region; the dashed vertical line between 2011 and 2012 marks the pre/post boundary relative to the heroin supply shock. Data are from the National Inpatient Sample (NIS), which covers years 2004–2019.

Figure A4. Event Study: Cocaine Overdose Admissions



Notes: This figure displays the β_τ coefficients from estimating Equation (1), capturing the event-time differences in the probability of inpatient admission for cocaine-related overdose between treatment regions (east of the Mississippi River) and control regions (west), relative to the omitted reference period (2011), which immediately precedes the supply shock. The event time spans from 2004 to 2019, excluding 2015 due to data unavailability. The regression includes fixed effects for Census division, calendar year and quarter, as well as individual-level demographic controls and discharge weights.

Figure A5. Heroin Source Area Distribution: 1977-2018



Source: DEA (2018b)